

ECEN 629 Convex Optimization for Electrical Engineering

Chapter 9 Unconstrained minimization

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Unconstrained minimization

$$\text{minimize } f(x)$$

- f convex, twice continuously differentiable (hence $\text{dom } f$ open)
- we assume optimal value $p^* = \inf_x f(x)$ is attained (and finite)

unconstrained minimization methods

- produce sequence of points $x^{(k)} \in \text{dom } f$, $k = 0, 1, \dots$ with

$$f(x^{(k)}) \rightarrow p^*$$

- can be interpreted as iterative methods for solving optimality condition

$$\nabla f(x^*) = 0$$

Initial point and sublevel set

algorithms in this chapter require a starting point $x^{(0)}$ such that

- $x^{(0)} \in \mathbf{dom} f$
- sublevel set $S = \{x \mid f(x) \leq f(x^{(0)})\}$ is closed

2nd condition is hard to verify, except when *all* sublevel sets are closed:

- equivalent to condition that $\mathbf{epi} f$ is closed
- true if $\mathbf{dom} f = \mathbf{R}^n$
- true if $f(x) \rightarrow \infty$ as $x \rightarrow \mathbf{bd} \mathbf{dom} f$

examples of differentiable functions with closed sublevel sets:

$$f(x) = \log\left(\sum_{i=1}^m \exp(a_i^T x + b_i)\right), \quad f(x) = -\sum_{i=1}^m \log(b_i - a_i^T x)$$

Strong convexity and implications

f is strongly convex on S if there exists an $m > 0$ such that

$$\nabla^2 f(x) \succeq mI \quad \text{for all } x \in S$$

implications

- for $x, y \in S$,

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{m}{2} \|x - y\|_2^2$$

hence, S is bounded

- $p^* > -\infty$, and for $x \in S$,

$$f(x) - p^* \leq \frac{1}{2m} \|\nabla f(x)\|_2^2$$

useful as stopping criterion (if you know m)

Descent methods

$$x^{(k+1)} = x^{(k)} + t^{(k)} \Delta x^{(k)} \quad \text{with } f(x^{(k+1)}) < f(x^{(k)})$$

- other notations: $x^+ = x + t\Delta x$, $x := x + t\Delta x$
- Δx is the *step*, or *search direction*; t is the *step size*, or *step length*
- from convexity, $f(x^+) < f(x)$ implies $\nabla f(x)^T \Delta x < 0$
(*i.e.*, Δx is a *descent direction*)

General descent method.

given a starting point $x \in \text{dom } f$.

repeat

1. Determine a descent direction Δx .
2. *Line search.* Choose a step size $t > 0$.
3. *Update.* $x := x + t\Delta x$.

until stopping criterion is satisfied.

Line search types

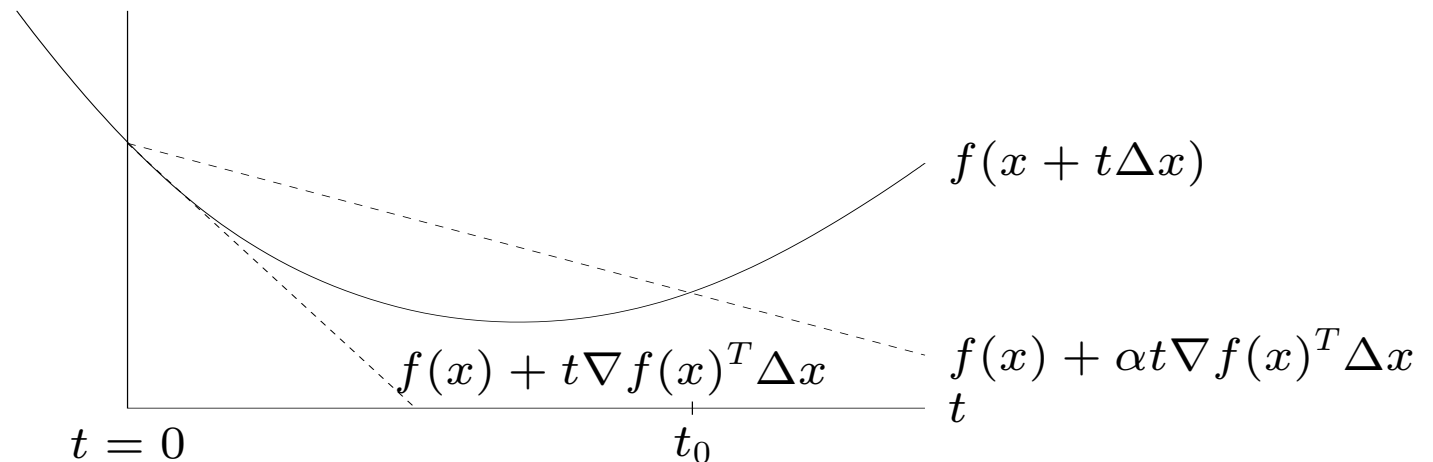
exact line search: $t = \operatorname{argmin}_{t>0} f(x + t\Delta x)$

backtracking line search (with parameters $\alpha \in (0, 1/2)$, $\beta \in (0, 1)$)

- starting at $t = 1$, repeat $t := \beta t$ until

$$f(x + t\Delta x) < f(x) + \alpha t \nabla f(x)^T \Delta x$$

- graphical interpretation: backtrack until $t \leq t_0$



Gradient descent method

general descent method with $\Delta x = -\nabla f(x)$

given a starting point $x \in \text{dom } f$.

repeat

1. $\Delta x := -\nabla f(x)$.
2. *Line search.* Choose step size t via exact or backtracking line search.
3. *Update.* $x := x + t\Delta x$.

until stopping criterion is satisfied.

- stopping criterion usually of the form $\|\nabla f(x)\|_2 \leq \epsilon$
- convergence result: for strongly convex f ,

$$f(x^{(k)}) - p^* \leq c^k (f(x^{(0)}) - p^*)$$

$c \in (0, 1)$ depends on m , $x^{(0)}$, line search type

- very simple, but often very slow; rarely used in practice

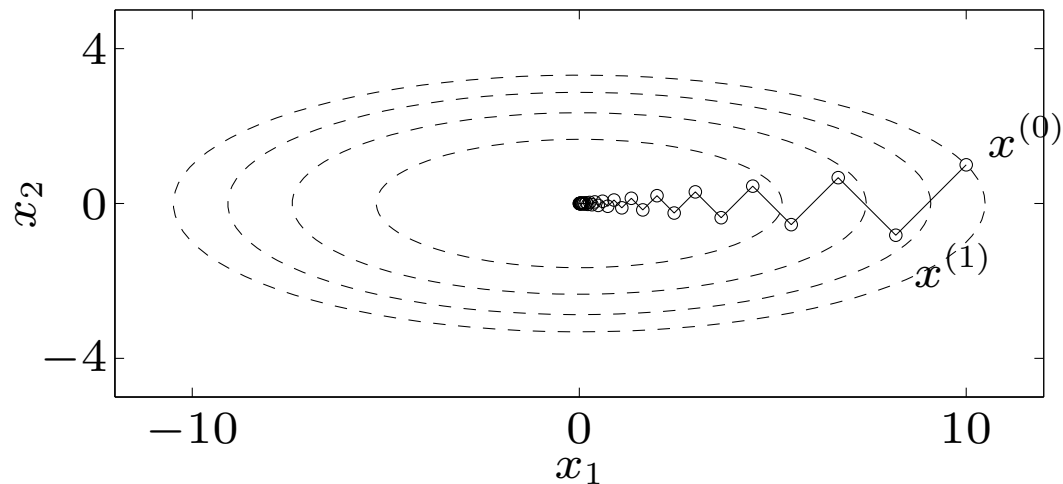
quadratic problem in $\mathbf{R}^2$

$$f(x) = (1/2)(x_1^2 + \gamma x_2^2) \quad (\gamma > 0)$$

with exact line search, starting at $x^{(0)} = (\gamma, 1)$:

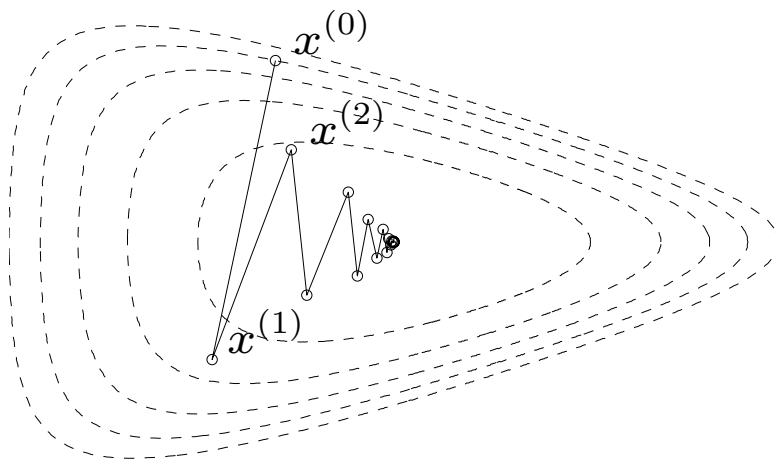
$$x_1^{(k)} = \gamma \left(\frac{\gamma - 1}{\gamma + 1} \right)^k, \quad x_2^{(k)} = \left(-\frac{\gamma - 1}{\gamma + 1} \right)^k$$

- very slow if $\gamma \gg 1$ or $\gamma \ll 1$
- example for $\gamma = 10$:

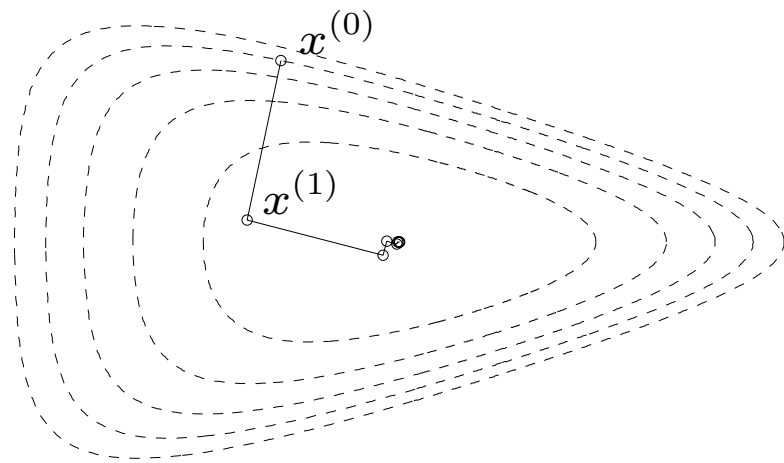


nonquadratic example

$$f(x_1, x_2) = e^{x_1+3x_2-0.1} + e^{x_1-3x_2-0.1} + e^{-x_1-0.1}$$



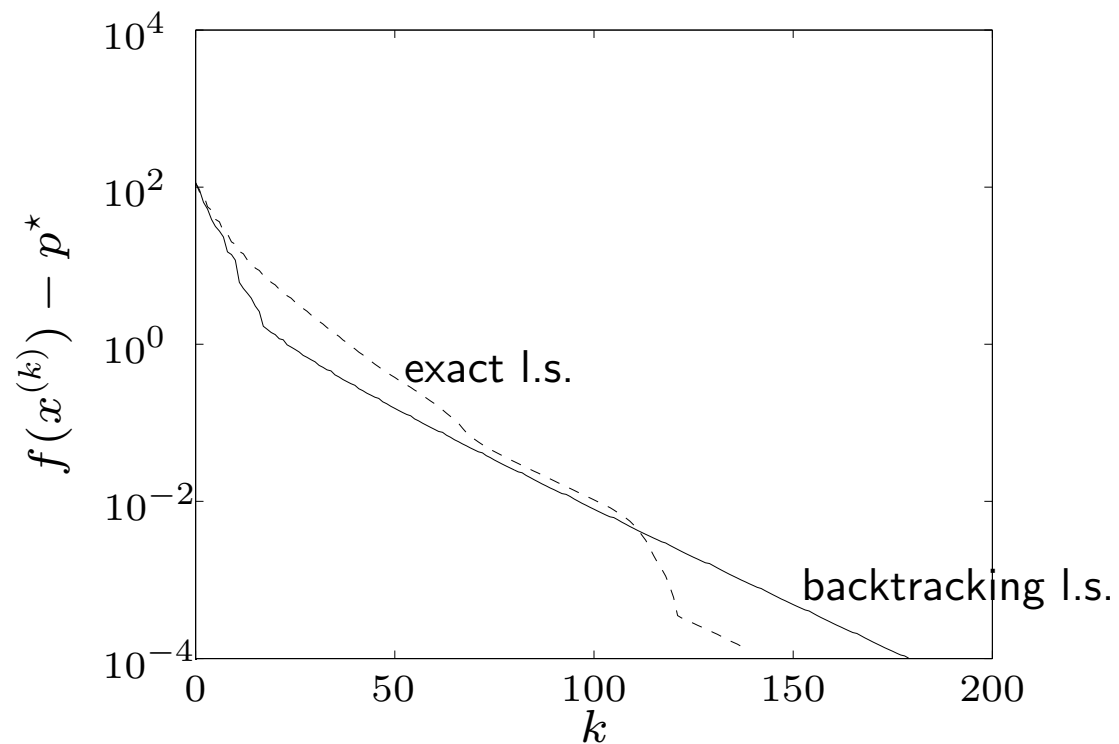
backtracking line search



exact line search

a problem in $\mathbf{R}^{100}$

$$f(x) = c^T x - \sum_{i=1}^{500} \log(b_i - a_i^T x)$$



‘linear’ convergence, *i.e.*, a straight line on a semilog plot

Steepest descent method: motivation

First order Taylor approximation of $f(x + v)$

$$f(x + v) \approx \hat{f}(x + v) = f(x) + \nabla f(x)^T v.$$

- Second term is the directional derivative;
- Approximate change in f for a small step v ;
- v is a step direction, toward reducing the function $f(x)$: normalization v .

Steepest descent method

normalized steepest descent direction (at x , for norm $\|\cdot\|$):

$$\Delta x_{\text{nsd}} = \operatorname{argmin}\{\nabla f(x)^T v \mid \|v\| = 1\}$$

interpretation: for small v , $f(x + v) \approx f(x) + \nabla f(x)^T v$;
direction Δx_{nsd} is unit-norm step with most negative directional derivative

(unnormalized) steepest descent direction

$$\Delta x_{\text{sd}} = \|\nabla f(x)\|_* \Delta x_{\text{nsd}}$$

satisfies $\nabla f(x)^T \Delta x_{\text{sd}} = -\|\nabla f(x)\|_*^2$

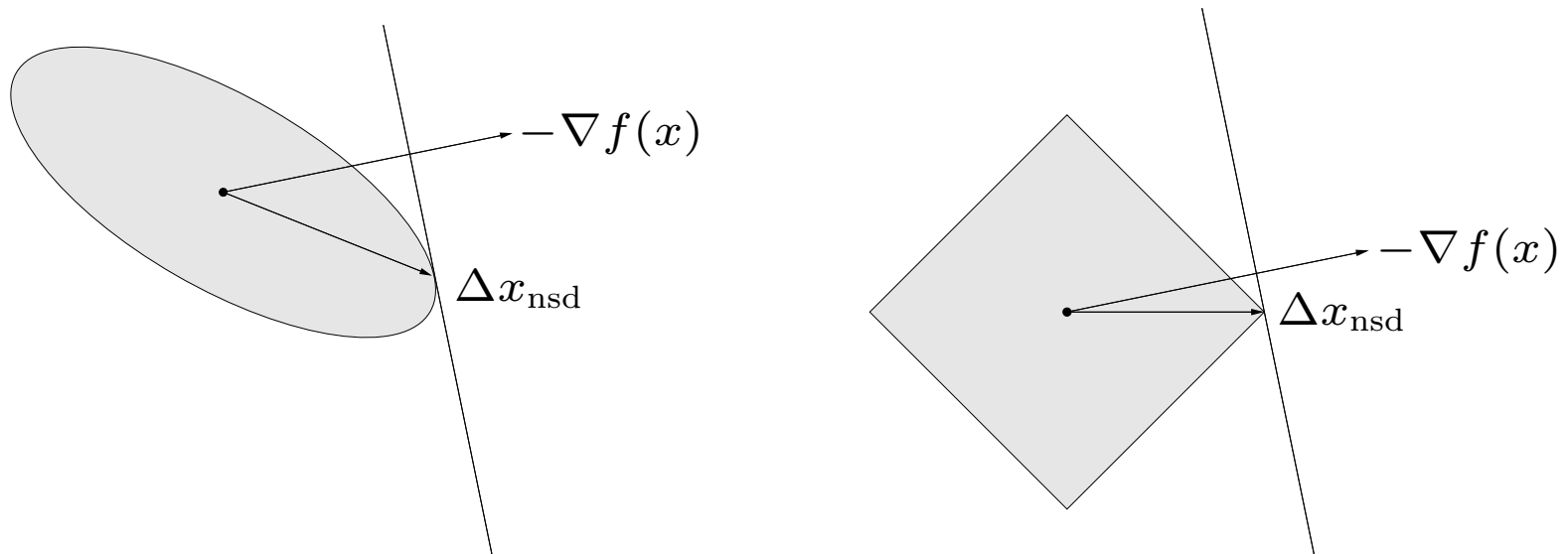
steepest descent method

- general descent method with $\Delta x = \Delta x_{\text{sd}}$
- convergence properties similar to gradient descent

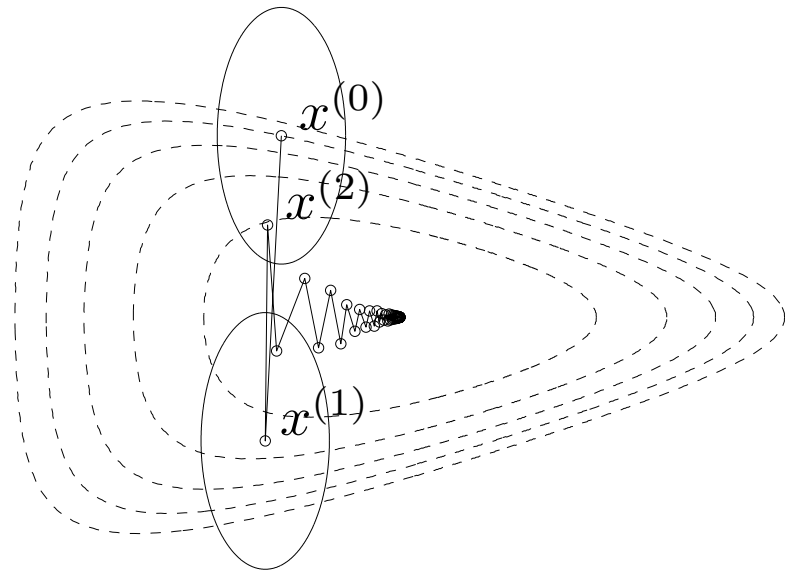
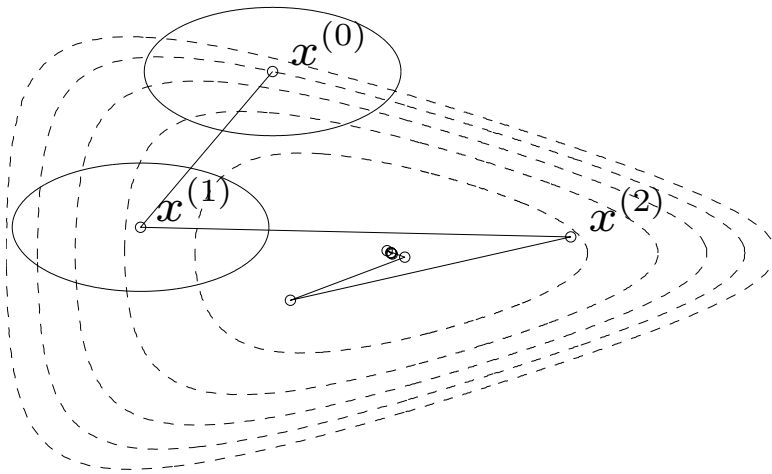
examples

- Euclidean norm: $\Delta x_{\text{sd}} = -\nabla f(x)$
- quadratic norm $\|x\|_P = (x^T P x)^{1/2}$ ($P \in \mathbf{S}_{++}^n$): $\Delta x_{\text{sd}} = -P^{-1} \nabla f(x)$
- ℓ_1 -norm: $\Delta x_{\text{sd}} = -(\partial f(x)/\partial x_i)e_i$, where $|\partial f(x)/\partial x_i| = \|\nabla f(x)\|_\infty$

unit balls and normalized steepest descent directions for a quadratic norm and the ℓ_1 -norm:



choice of norm for steepest descent



- steepest descent with backtracking line search for two quadratic norms
- ellipses show $\{x \mid \|x - x^{(k)}\|_P = 1\}$
- equivalent interpretation of steepest descent with quadratic norm $\|\cdot\|_P$: gradient descent after change of variables $\bar{x} = P^{1/2}x$

shows choice of P has strong effect on speed of convergence

Steepest descent with $\|\cdot\|_1$ norm

The solutions is (why?)

$$\Delta x_{\text{nsd}} = -\text{sign} \left(\frac{\partial f(x)}{\partial x_i} \right) e_i,$$

where x_i is the coordinate that has the largest partial derivative component.

- Choose this coordinate, then decrease or increase it;
- Each iteration only changes one coordinate: a coordinate-descent algorithm;
- The line search can be simplified.

Example: Frobenius norm scaling

Consider the geometric program

$$\text{minimize: } \sum_{i,j=1}^n M_{ij}^2 d_i^2 / d_j^2,$$

where $M \in \mathbb{R}^{n \times n}$ is given and $d \in \mathbb{R}^n$ is the variable. Equivalent to

$$\text{minimize: } f(x) = \log \left(\sum_{i,j=1}^n M_{ij}^2 e^{x_i - x_j} \right),$$

by substituting $x_i = 2 \log d_i$. Consider coordinate descent by isolating x_k

$$f(x) = \log(\alpha_k + \beta_k e^{-x_k} + \gamma_k e^{x_k}).$$

This can be solved explicitly $x_k = \log(\beta_k / \gamma_k) / 2$. $\Rightarrow$ Algorithm?

More about Descent Algorithms

In theoretical analysis of descent algorithms with constraints, two approaches are often taken:

- Projected descent: first $y^+ = x + \eta \Delta x$, then the real choice of x^+ is the projection of y^+ onto the feasible set;
- Frank-Wolfe step: first find $y^+ = \arg \min \Delta x^T y$, then $x^+ = (1 - \eta)x + \eta y^+$.

Note: both take into consideration the constraints naturally, though not efficiently: only relies on the abstract version of the constraint set.

Newton step

$$\Delta x_{\text{nt}} = -\nabla^2 f(x)^{-1} \nabla f(x)$$

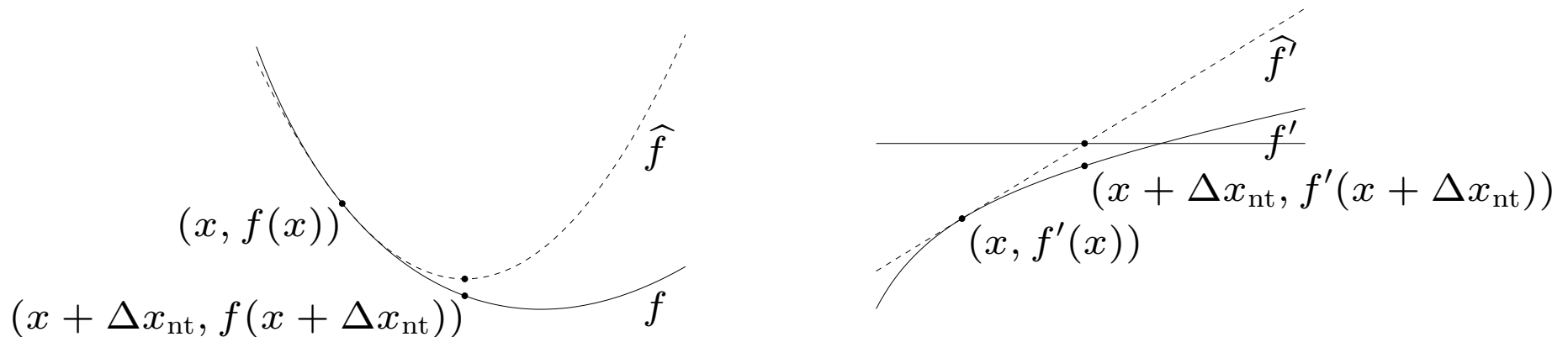
interpretations

- $x + \Delta x_{\text{nt}}$ minimizes second order approximation

$$\hat{f}(x + v) = f(x) + \nabla f(x)^T v + \frac{1}{2} v^T \nabla^2 f(x) v$$

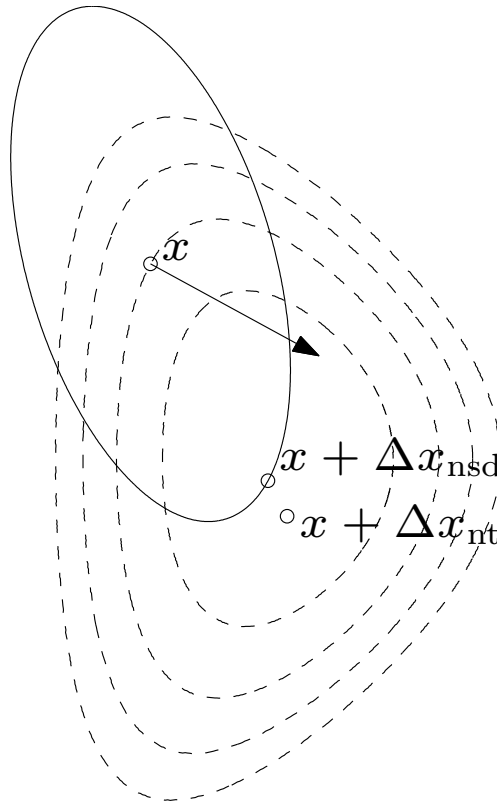
- $x + \Delta x_{\text{nt}}$ solves linearized optimality condition

$$\nabla f(x + v) \approx \nabla \hat{f}(x + v) = \nabla f(x) + \nabla^2 f(x) v = 0$$



- Δx_{nt} is steepest descent direction at x in local Hessian norm

$$\|u\|_{\nabla^2 f(x)} = (u^T \nabla^2 f(x) u)^{1/2}$$



dashed lines are contour lines of f ; ellipse is $\{x + v \mid v^T \nabla^2 f(x) v = 1\}$

arrow shows $-\nabla f(x)$

Newton decrement

$$\lambda(x) = \left(\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x) \right)^{1/2}$$

a measure of the proximity of x to x^*

properties

- gives an estimate of $f(x) - p^*$, using quadratic approximation $\hat{f}$:

$$f(x) - \inf_y \hat{f}(y) = \frac{1}{2} \lambda(x)^2$$

- equal to the norm of the Newton step in the quadratic Hessian norm

$$\lambda(x) = \left(\Delta x_{\text{nt}}^T \nabla^2 f(x) \Delta x_{\text{nt}} \right)^{1/2}$$

- directional derivative in the Newton direction: $\nabla f(x)^T \Delta x_{\text{nt}} = -\lambda(x)^2$
- affine invariant (unlike $\|\nabla f(x)\|_2$)

Newton's method

given a starting point $x \in \text{dom } f$, tolerance $\epsilon > 0$.

repeat

1. *Compute the Newton step and decrement.*

$$\Delta x_{\text{nt}} := -\nabla^2 f(x)^{-1} \nabla f(x); \quad \lambda^2 := \nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x).$$

2. *Stopping criterion.* **quit** if $\lambda^2/2 \leq \epsilon$.

3. *Line search.* Choose step size t by backtracking line search.

4. *Update.* $x := x + t\Delta x_{\text{nt}}$.

affine invariant, *i.e.*, independent of linear changes of coordinates:

Newton iterates for $\tilde{f}(y) = f(Ty)$ with starting point $y^{(0)} = T^{-1}x^{(0)}$ are

$$y^{(k)} = T^{-1}x^{(k)}$$

Classical convergence analysis

assumptions

- f strongly convex on S with constant m
- $\nabla^2 f$ is Lipschitz continuous on S , with constant $L > 0$:

$$\|\nabla^2 f(x) - \nabla^2 f(y)\|_2 \leq L\|x - y\|_2$$

(L measures how well f can be approximated by a quadratic function)

outline: there exist constants $\eta \in (0, m^2/L)$, $\gamma > 0$ such that

- if $\|\nabla f(x)\|_2 \geq \eta$, then $f(x^{(k+1)}) - f(x^{(k)}) \leq -\gamma$
- if $\|\nabla f(x)\|_2 < \eta$, then

$$\frac{L}{2m^2} \|\nabla f(x^{(k+1)})\|_2 \leq \left(\frac{L}{2m^2} \|\nabla f(x^{(k)})\|_2 \right)^2$$

damped Newton phase ($\|\nabla f(x)\|_2 \geq \eta$)

- most iterations require backtracking steps
- function value decreases by at least γ
- if $p^* > -\infty$, this phase ends after at most $(f(x^{(0)}) - p^*)/\gamma$ iterations

quadratically convergent phase ($\|\nabla f(x)\|_2 < \eta$)

- all iterations use step size $t = 1$
- $\|\nabla f(x)\|_2$ converges to zero quadratically: if $\|\nabla f(x^{(k)})\|_2 < \eta$, then

$$\frac{L}{2m^2} \|\nabla f(x^l)\|_2 \leq \left(\frac{L}{2m^2} \|\nabla f(x^k)\|_2 \right)^{2^{l-k}} \leq \left(\frac{1}{2} \right)^{2^{l-k}}, \quad l \geq k$$

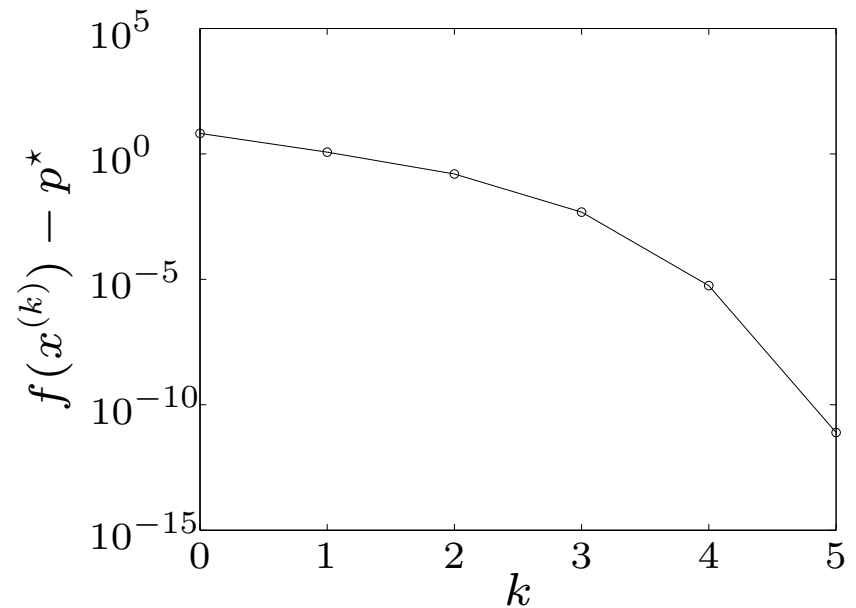
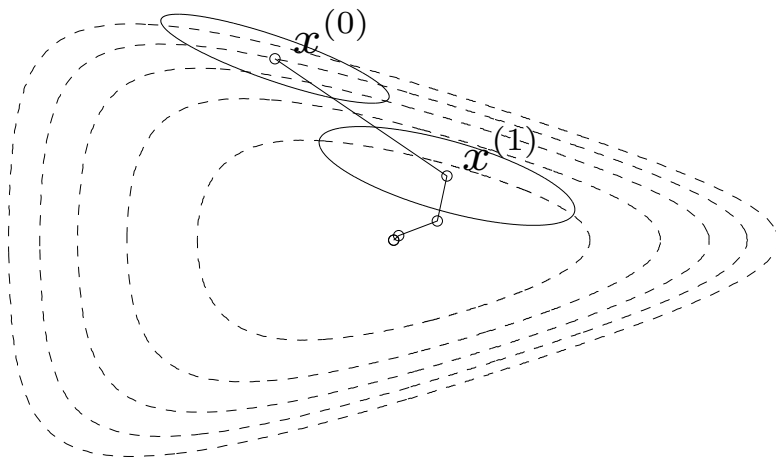
conclusion: number of iterations until $f(x) - p^* \leq \epsilon$ is bounded above by

$$\frac{f(x^{(0)}) - p^*}{\gamma} + \log_2 \log_2(\epsilon_0/\epsilon)$$

- γ, ϵ_0 are constants that depend on $m, L, x^{(0)}$
- second term is small (of the order of 6) and almost constant for practical purposes
- in practice, constants m, L (hence γ, ϵ_0) are usually unknown
- provides qualitative insight in convergence properties (*i.e.*, explains two algorithm phases)

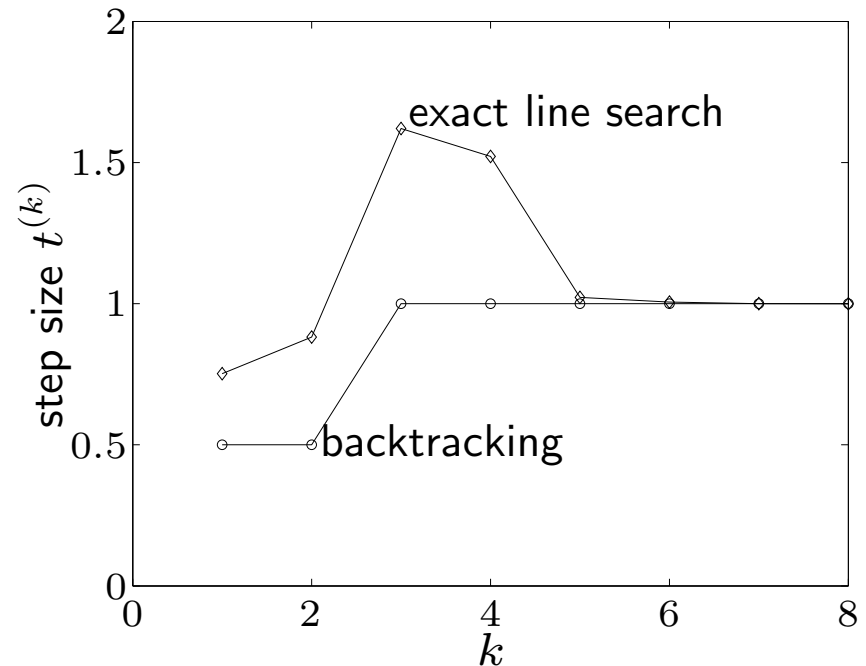
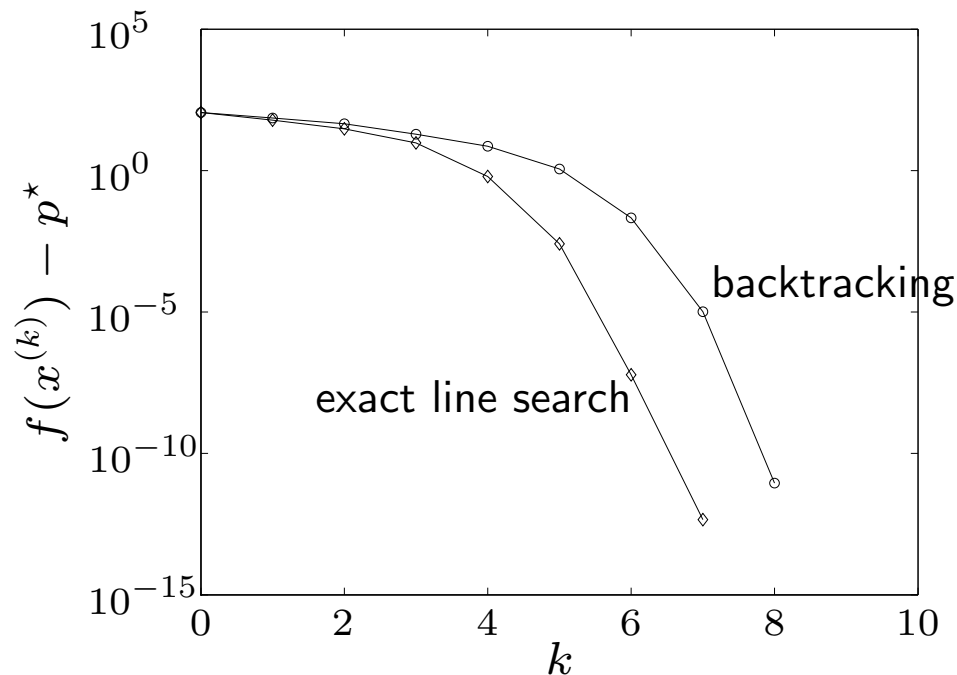
Examples

example in $\mathbf{R}^2$ (page 10–9)



- backtracking parameters $\alpha = 0.1$, $\beta = 0.7$
- converges in only 5 steps
- quadratic local convergence

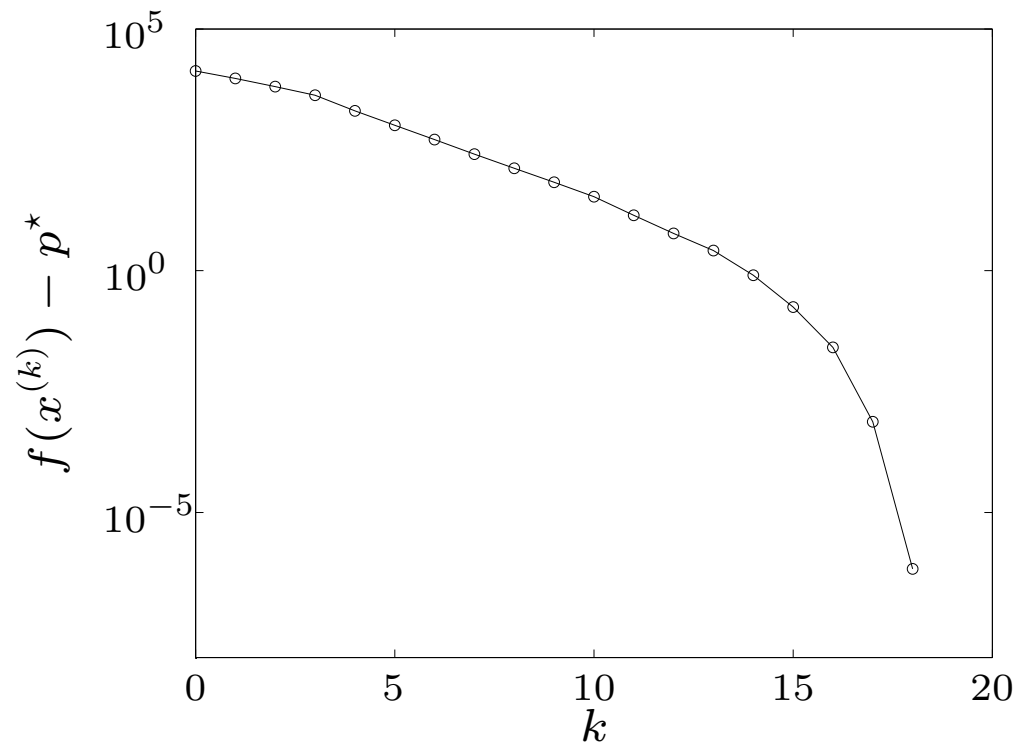
example in $\mathbf{R}^{100}$ (page 10–10)



- backtracking parameters $\alpha = 0.01$, $\beta = 0.5$
- backtracking line search almost as fast as exact l.s. (and much simpler)
- clearly shows two phases in algorithm

example in $\mathbf{R}^{10000}$ (with sparse a_i)

$$f(x) = - \sum_{i=1}^{10000} \log(1 - x_i^2) - \sum_{i=1}^{100000} \log(b_i - a_i^T x)$$



- backtracking parameters $\alpha = 0.01$, $\beta = 0.5$.
- performance similar as for small examples

Summary on Newton's method

Strong advantages over gradient and steepest descent

- Rapid convergence, particularly, quadratic near x^* ;
- Affine invariant: insensitive to change of coordinates or linear transforms;
- Scale well with problem sizes;
- Not dependent on algorithm parameters (e.g., backtracking parameters).

Summary on Newton's method

Strong advantages over gradient and steepest descent

- Rapid convergence, particularly, quadratic near x^* ;
- Affine invariant: insensitive to change of coordinates or linear transforms;
- Scale well with problem sizes;
- Not dependent on algorithm parameters (e.g., backtracking parameters).

Disadvantages

- Cost of forming and storing the Hessian (matrix);
- The cost of computing the Newton step (solving linear equations);
- Quasi-Newton method: approximately solve the Newton step, sometimes have the same advantages of the Newton method.

Exercise 1

The quadratic problem

$$\text{minimize: } \frac{1}{2}(x_1^2 + \gamma x_2^2)$$

where $\gamma > 0$.

For the current point $x = (x_1, x_2)$, e.g., $x = (2, 8)$, and $(\alpha, \beta, \gamma) = (0.45, 0.9, 0.1)$:

- Find the gradient, find the corresponding backtracking line search stepsize for the gradient descent algorithm;
- Find the steepest descent direction using ℓ_1 norm and ℓ_∞ norm, and the corresponding backtracking line search stepsizes;
- Find the Newton step, the corresponding backtracking line search stepsize, and the Newton decrement.

Exercise 2

A more general problem

$$\text{minimize: } \exp(2x_1 + x_2) + \frac{1}{2}(x_1^2 + \gamma x_2^2)$$

where $\gamma > 0$.

For the current point $x = (x_1, x_2)$:

- Find the gradient, find the corresponding backtracking line search stepsize for the gradient descent algorithm;
- Find the steepest descent direction using ℓ_1 norm and ℓ_∞ norm, and the corresponding backtracking line search stepsizes;
- Find the Newton step, the corresponding backtracking line search stepsize, and the Newton decrement.